

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester VI Examination, 2022-23

Paper No. BPT-604: Atomic Physics and Lasers

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer all questions. Symbols have their usual meanings.***Question 1.** Answer any **five** of the following:

[5 × 4 = 20]

- (a) A muonic atom contains a nucleus of one proton and a negative muon μ^- with charge $-e$ and a mass 207 times as large as an electron mass. What is the wavelength of the first line in the Lyman series for such an atom?
- (b) In a Franck-Hertz type of experiment, atomic hydrogen is bombarded with electrons and excitation potentials are found at 10.20 V and 12.10 V.
 - (i) Draw an energy-level diagram and show the possible spectral emissions accompanying these excitations.
 - (ii) What are the wavelengths of the emitted photons?
- (c) Show that the selection rules for orbital angular momentum and orbital magnetic quantum number are $\Delta l = \pm 1$ and $\Delta m_l = 0, \pm 1$.
- (d) A beam of hydrogen atoms emitted from an oven at temperature 400 K is sent through a Stern-Gerlach magnet of length 1 m. The transverse magnetic field gradient is 10 T/m. Calculate the transverse deflection of the components of the beam at the point where the beam leaves the magnet. Take kinetic energy as $2kT$.
- (e) Show the suitability of a two-level atomic system for amplification of radiation in microwave and optical frequency regions.
- (f) Evaluate the ratio of the orbital magnetic dipole moment to the orbital angular momentum, μ_l/L , for an electron moving in an elliptical orbit of the Bohr-Sommerfeld atom. Compare the results with those of a circular orbit.
- (g) Consider the states in which $l = 4$ and $s = 1/2$. For the state with the largest possible j and largest possible m_j , calculate: (i) the angle between L and S , (ii) the angle between μ_L and μ_S , and (iii) the angle between J and the $+z$ axis.

Question 2.

- (a) The spin-orbit interaction energy is $\Delta E = \frac{1}{2m^2c^2} \frac{1}{r} \frac{dV(r)}{dr} \vec{S} \cdot \vec{L}$ and the term shift due to relativistic correction of mass for H-atom is $\Delta T_r = \frac{R\alpha^2 Z^4}{n^3} \left[\frac{1}{l+1/2} - \frac{3}{4n} \right]$. Discuss the fine structure of the H_α line showing clearly the splitting and possible transitions. [8.5]
- (b) Compute the ground state spectroscopic terms of carbon and oxygen atoms. How are normal and inverted multiplets different? [4]

Question 3.

- (a) What is the Zeeman effect? Derive the expression for interaction energy in a magnetic field much less than the internal magnetic field and apply it to find the components of sodium D_1 and D_2 lines. [8.5]
- (b) Compute the intensity ratio of the three components of the first line of the diffuse series. [4]
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Question 4.

- (a) Discuss the splitting of $^2P_{3/2}$ and $^2S_{1/2}$ states of sodium atom in a strong magnetic field (greater than the internal magnetic field) and show the transitions. Is it the same as the normal Zeeman effect? [6.5]
- (b) In an atom obeying L-S coupling, there are two normal triplets. The separations between adjacent components of the lower triplet are 20 cm^{-1} and 40 cm^{-1} , and for the higher state triplet the separations are 22 cm^{-1} and 33 cm^{-1} . Determine the terms for the two triplet states and show the allowed transitions with the help of an energy level diagram. [6]
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Question 5.

- (a) What are the Einstein coefficients for absorption and emission processes between two levels? Compute the relation between them. [8]
- (b) What do you mean by molecular spectra? Explain in brief. [4.5]

OR

- (a) Discuss the construction and working of a He-Ne Laser. With an energy level diagram, explain the reason for the narrow bore of the Laser tube. [8]
- (b) Discuss the rate equations for a three-level Laser system. [4.5]